

Homework Assignment 2

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Problem 0.1. Let $U \subseteq \mathbb{C}$ be open. Prove that ∂_z and $\partial_{\bar{z}}$ are derivations on U .

Proof. This proof is inspired by the fact that the collection of vector fields $\mathfrak{X}(M)$ on a manifold M is a vector space. In this case, the manifold would need to be complex such that $\mathfrak{X}(M)$ would be a $\mathbb{C}$ -vector space.

Let F, G be derivations on U , and let $aF + bG$, $a, b \in \mathbb{C}$ be a $\mathbb{C}$ -linear combination of derivations. We claim that this is also a derivation. Linearity of $aF + bG$ follows by the usual proof: given $f, g \in C^1(U, \mathbb{C})$ and $w, z \in \mathbb{C}$,

$$\begin{aligned}(aF + bG)(w + z) &= aF(wf + zg) + bG(wf + zg) \\ &= aF(wf) + aF(zg) + bG(wf) + bG(zg) \\ &= waF(f) + zaF(g) + wbG(f) + zbG(g) \\ &= w(aF + bG)(f) + z(aF + bG)(g).\end{aligned}$$

To prove the product rule, we appeal to the fact that F, G are derivations. We have

$$\begin{aligned}(aF + bG)(fg) &= aF(fg) + bG(fg) \\ &= a(fF(g) + gF(f)) + b(fG(g) + gG(f)) \\ &= f(aF + bG)(g) + g(aF + bG)(f).\end{aligned}$$

Thus $aF + bG$ is a derivation. Since ∂_x, ∂_y are $\mathbb{C}$ -linear they are derivations on U . ■

Problem 0.2. Prove that if f is holomorphic on $\mathbb{C}$ and real-valued, then f is constant.

Proof. Let $f = u + iv$ for $u, v : \mathbb{C} \rightarrow \mathbb{R}$. Since f is real-valued, $v = 0$, so $\partial_x v = \partial_y v = 0$. The Cauchy-Riemann equations then read as follows:

$$\frac{\partial u}{\partial x} = 0, \quad \frac{\partial u}{\partial y} = 0.$$

Since $f = u$, we thus have that f has vanishing partials. Constant functions are precisely those functions whose partials all vanish. Thus f is constant. ■

Problem 0.3. Prove that if

$$L = a \frac{\partial}{\partial x} + b \frac{\partial}{\partial y}$$

and

$$M = c \frac{\partial}{\partial x} + d \frac{\partial}{\partial y}$$

(with $a, b, c, d \in \mathbb{C}$) and if

$$\begin{aligned}Lz &= 1, & L\bar{z} &= 0, \\ Mz &= 0, & M\bar{z} &= 1,\end{aligned}$$

then it must be that $L = \partial_z$ and $M = \partial_{\bar{z}}$.

Proof. Let notation and assumptions be as given in the problem, with $z = x + iy$. By assumption, $Lz = 1$ implies that

$$\begin{aligned} 1 &= (a\partial_x + b\partial_y)(x + iy) \\ &= a\partial_x(x + iy) + b\partial_y(x + iy) \\ &= a + ib. \end{aligned}$$

This does not imply $b = 0$ since $\{1, i\}$ is not $\mathbb{C}$ -linearly independent. However, applying $L\bar{z} = 0$ yields

$$\begin{aligned} 0 &= (a\partial_x + b\partial_y)(x - iy) \\ &= a\partial_x(x - iy) + b\partial_y(x - iy) \\ &= a - ib. \end{aligned}$$

We obtain $a = ib$, and plugging into $a + ib = 1$ yields $2a = 1$. Thus $a = \frac{1}{2}$ and

$$ib = \frac{1}{2} \implies b = \frac{1}{2i} = -\frac{i}{2}.$$

Thus

$$L = \frac{1}{2} \frac{\partial}{\partial x} - \frac{i}{2} \frac{\partial}{\partial y} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) = \frac{\partial}{\partial z}.$$

We apply the same process to M .

$$\begin{aligned} 0 &= (c\partial_x + d\partial_y)(x + iy) \\ &= c + id \\ 1 &= (c\partial_x + d\partial_y)(x - iy) \\ &= c - id. \end{aligned}$$

Thus $c = -id$ implying that $1 = -2id$, and thus $d = \frac{i}{2}$. We then obtain $c = \frac{1}{2}$, yielding

$$M = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) = \frac{\partial}{\partial \bar{z}}.$$

■

Problem 0.4. If f is a C^1 function on an open set $U \subseteq \mathbb{C}$, then prove that

$$\overline{\left(\frac{\partial f}{\partial z} \right)} = \frac{\partial \bar{f}}{\partial \bar{z}}.$$

Proof. This is once again a straightforward computation. Recall that all complex functions $f : U \rightarrow \mathbb{C}$ are of the form $f = u + iv$ for $u, v : U \rightarrow \mathbb{R}$. We then obtain

$$\begin{aligned} \overline{\left(\frac{\partial f}{\partial z} \right)} &= \frac{1}{2} \overline{\left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} - i \frac{\partial u}{\partial y} + i^2 \frac{\partial v}{\partial y} \right)} \\ &= \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} + i \left(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \right) \right) \\ \frac{\partial \bar{f}}{\partial \bar{z}} &= \frac{\partial}{\partial \bar{z}}(u - iv) \\ &= \frac{1}{2} \left(\frac{\partial u}{\partial x} - i \frac{\partial v}{\partial x} + i \frac{\partial u}{\partial y} + i^2 \frac{\partial v}{\partial y} \right) \\ &= \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} + i \left(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \right) \right). \end{aligned}$$

The computations agree.

